

Busco $\int_0^1 B_\nu(x) dx.$

Propiedad: $\mathcal{D}_x [B_\nu^\wedge(x)] = \mathcal{D} [B_\nu^\wedge(x)] = \nu B_{\nu-1}^\wedge(x)$

Luego:

$$\mathcal{D}(B_{\nu+1}(x)) = \sqrt{\nu+1} B_\nu(x)$$

Integro y divido por $\nu+1$ n.s.m:

$$\int \frac{\mathcal{D} B_{\nu+1}(x)}{\nu+1} dx = \int B_\nu(x) dx$$

$$\frac{1}{\nu+1} B_{\nu+1}(x) + C = \int B_\nu(x) dx$$

Entonces:

$$\int_0^1 B_\nu(x) dx = \frac{1}{\nu+1} B_{\nu+1}(x) \Big|_{x=0}^{x=1} = \frac{1}{\nu+1} (B_{\nu+1}(1) - B_{\nu+1}(0))$$

$$\int_0^1 B_\nu(x) dx = \frac{1}{\nu+1} \Delta B_{\nu+1}(x) \Big|_{x=0}$$

Propiedad: $\Delta B_\nu^\wedge(x) = \nu B_{\nu-1}^\wedge(x)$

Luego:

$$\int_0^1 B_\nu(x) dx = \frac{1}{\nu+1} (\nu+1) B_\nu^\circ(x) \Big|_{x=0}$$

Propiedad: $B_v^0(x) = x^v$

Luego: $\int_0^1 B_v(x) dx = B_v^0(1) = 1^v = 1$ (si $v \neq 0$)

Propiedad $B_0(x) = 1 \Rightarrow \int_0^1 B_0(x) dx = \int_0^1 1 dx = x \Big|_0^1 = 1$

Luego:

$$\int_0^1 B_v(x) dx = \begin{cases} 0 & v \neq 0 \\ 1 & v = 0 \end{cases}$$